

Cloud Computing TASK – 3

- 1. Pick any two apps you use daily (like Zomato, Instagram, or Paytm) and identify which cloud deployment model (Public, Private, Hybrid, or Community) would best suit each app if you were the CTO. Justify your choice with two reasons for each.**

Ans :

Application	Suitable Cloud Deployment Model	Justification
Instagram	Public Cloud	1. Public cloud provides unlimited scalability, which helps Instagram handle millions of users uploading photos, videos, and stories every day. 2. It reduces infrastructure costs because resources can be increased or decreased based on user demand.
Paytm	Hybrid Cloud	1. Hybrid cloud allows Paytm to keep sensitive financial data and transactions in a private environment for better security. 2. It can use public cloud resources during high-demand periods, such as festivals and

Application	Suitable Cloud Deployment Model	Justification
		sales, to handle sudden increases in transactions.

2. Create a comparison table listing the pros and cons of Public, Private, Hybrid, and Community cloud deployment models. Include at least three points each for pros and cons.

Ans :

Cloud Model	Pros	Cons
Public Cloud	1. Low cost because resources are shared among multiple users. 2. Highly scalable and can handle increasing workloads easily. 3. No need to manage physical hardware; maintenance is handled by the provider.	1. Less control over infrastructure compared to private cloud. 2. Security concerns may arise due to shared resources. 3. Performance can depend on provider availability and network conditions.
Private Cloud	1. Provides high security and privacy for sensitive data. 2. Offers complete control over	1. Higher cost due to dedicated hardware and maintenance. 2. Requires skilled IT staff for management.

Cloud Model	Pros	Cons
	infrastructure and resources. 3. Can be customized according to an organization's specific needs.	3. Scaling resources can be slower and more expensive.
	1. Combines the benefits of both public and private clouds. 2. Allows sensitive data to remain private while using public cloud scalability. 3. Provides flexibility for handling changing workloads.	1. More complex to manage and maintain. 2. Integration between private and public environments can be challenging. 3. Requires proper security management across multiple environments.
Hybrid Cloud	1. Shared by organizations with similar goals or requirements. 2. Provides better security and compliance than public cloud for specific groups. 3. Costs can be shared among participating organizations.	1. Limited scalability compared to public cloud. 2. Requires agreement and coordination between community members. 3. Higher management complexity due to shared ownership.
Community Cloud		

- 3. Read the case study of WhatsApp's cloud infrastructure (search online for details) and summarize which deployment model they use, explaining why this model is suitable for their business needs.

Hint: Focus on scalability, security, and global reach.**

Ans :

WhatsApp mainly uses a Public Cloud deployment model as part of Meta's cloud infrastructure. It uses large-scale cloud data centers and distributed systems to provide messaging services to billions of users worldwide.

Why This Model Is Suitable for WhatsApp

1. Scalability:

Public cloud allows WhatsApp to handle a huge number of users, messages, photos, videos, and calls by increasing computing resources when demand grows.

2. Security:

WhatsApp uses strong security features such as end-to-end encryption and additional protections for user data, helping maintain privacy while operating at a global scale.

3. Global Reach and Availability:

Using a large cloud infrastructure allows WhatsApp to provide fast and reliable services to users across different countries with high availability.

- 4. Imagine you are launching a new online ticket booking platform for IPL matches. Choose a cloud deployment model for your platform, list two possible risks with your choice, and suggest a way to mitigate one of those risks.**

Ans :

Selected Cloud Deployment Model: Hybrid Cloud

For an online IPL ticket booking platform, I would choose a Hybrid Cloud deployment model. It combines the security of a private cloud with the scalability of a public cloud.

Possible Risks:

1. Complex Management:

Managing both private and public cloud environments can be complicated and requires skilled IT professionals.

2. Security Risks During Data Transfer:

Moving data between private and public cloud environments may create security vulnerabilities if not properly protected.

Risk Mitigation:

To reduce security risks, the platform can use strong encryption, secure network connections (such as VPNs), and proper access controls while transferring data between cloud environments.